Ryan Nguyen

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Education

University of California, Berkeley **Expected Graduation: December 2027**

Bachelor's in Computer Science

- **Scholarship Foundation Exceptional Leadership Scholarship** Recipient
- Relevant Coursework: Data Structures & Algorithms, Efficient Algorithms & Intractable Problems, Computer Architecture, Operating Systems, Machine Learning, Databases, Discrete Mathematics & Probability
- Campus Involvement: **Web Development @ Berkeley** (President of Boot-camp Development & Industry Developer)

Experience

Software Development Engineering Intern, Amazon Web Services **Aug 2026 - Present**

- Building automated pipeline to detect and integrate **open-source SMT solver** releases into automated reasoning systems.
- Engineering end-to-end automation to **build and containerize SMT solvers**, benchmark new releases against relevant prior versions for **soundness and regressions**, and mechanically standardize solver inputs for reproducible testing.

Software Engineering Intern, Genuine Parts Company **Jun 2026 - Aug 2026**

- Built a 360° view customer intelligence platform and **Elasticsearch** service integrating cross-domain data aggregation.
- Rearchitected the data retrieval pipeline, parallelizing **14+ independent BigQuery workloads** with multithreading and asynchronous execution, reducing end-to-end search latency by **86% (23s to 3s)**.
- Designed multi-tier Redis caching layer, accelerating lookups from **3s→100ms**, reducing BigQuery compute costs by **80%**.
- Optimized frontend performance through concurrent data fetching, progressive rendering, hover prefetching, and compressed API responses, reducing time-to-first-content by **66% (3s to 2s)** and **eliminating blocking UI behavior**.

Technical Lead, LearnHaus AI **Jan 2026 - May 2026**

- Shipped the company's first **LLM evaluation platform**, leading a team of **9 engineers**, standardizing benchmarking across **4+ frontier models** and unblocking model selection, regression testing, and prompt iteration for ML and product teams.
- Designed and shipped three integrated workflows—scoring playground, human-vs-model comparison, and dataset-level pattern analysis—with unified side-by-side diffing, compressing iteration cycles from **2–3 days to <2 hours (95% reduction)**.
- Engineered statistical diffing and analytics tooling **(MAE, variance, correlation, clustering**) that surfaced **12+ scoring biases** across **1K+ production outputs**, directly informing model retraining and evaluation priorities.

Teaching Assistant, Full-Stack Development - UC Berkeley [Link] **Nov 2025 - May 2026**

- Instructed **100+ students/semester** across lectures and office hours on **React, Node/Express, Flask/Django, MongoDB, SQL, Firebase, and DevOps**, supporting debugging and full-stack project development.
- Engineered course infrastructure at scale: authored **20+ pytest/Jest autograders** for React components, REST endpoints, and auth flows; submission platform for **2000+ assignments**; shipped a ticketing system and **15+ automation scripts**.

Projects

Volare: Review & Directory Service [Link]| *React, TypeScript, Spring Boot, WebFlux, Resilience4j, Docker*

- Built and deployed a **full-stack AI travel platform** for daily itineraries, real-time flight search, and restaurant discovery.
- Engineered a reactive backend with **WebFlux, Resilience4j circuit breakers/retries, and async caching**, reducing cache-hit latency by **99% (1ms vs. 400–600ms upstream)**, **95% less redundant API calls** during upstream failures.
- Diagnosed and resolved a production-critical Spring WebClient shared-state auth leak causing cascading 401 failures across all 3 API integrations; automated CI/CD and multi-service deployment with Docker, GitHub Actions, Render, and Vercel

LockIn [Link]| *JavaScript, Chrome MV3, MutationObserver, CSS* `:has()`

- Built a lightweight **5.5KB Chrome extension** with **zero dependencies** that removes YouTube Shorts and Instagram Reels across **5 layout variants** (including A/B tests) using a **debounced MutationObserver**.
- Pivoted from CSS `:has()` to a JS attribute-stamping hide system after profiling render jank on YouTube's virtualized feed; **7 hand-tuned selectors** target every known Shorts and Reels DOM variant with **zero false positives**.

HTTP Reverse Proxy + Load Balancer [Link]| *Go,* `net/http/httputil`

- Engineered a **zero-dependency HTTP reverse proxy and load balancer** in Go with round-robin, least-connections, and IP-hash sticky routing across **per-route backend pools** with **longest-prefix path matching** for deterministic routing.
- Engineered fault tolerance via health-checked backend eviction/re-admission and **bounded retry-on-5xx across up to 3 backend replicas**, enabling seamless failover and masking upstream failures for idempotent requests.
- Implemented **TLS termination**, structured logging, and graceful shutdown with in-flight request draining.

Technical Skills

Languages: Python, Java, Go, C/C++, JavaScript, TypeScript, HTML/CSS(Tailwind, BOLT), SQL, RISC-V Assembly
Backend: Node.js, Express.js, FastAPI, Flask, Django, Spring Boot (Data JPA, WebClient, WebFlux), REST APIs, PostgreSQL, MySQL, MongoDB, Firebase/Firestore, AWS, Google Cloud (BigQuery, Vertex AI) Redis, Elasticsearch
Frontend & Systems: React, Next.js, Chrome Extensions (MV3), TensorFlow, PyTorch, Docker, CI/CD, Git, Bash, Linux, Render, Vercel, pytest, Jest, Vite